

Bhargav Koushal

Srinagar, J&K, India

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Summary

Software Engineering student at NIT Srinagar with hands-on experience in Generative AI, Machine Learning, NLP, full-stack development, and backend systems. Experienced in building AI-powered applications, ML models, REST APIs, research-oriented NLP systems, and web applications using Python and modern development frameworks.

Education

National Institute of Technology Srinagar <i>B.Tech in Information Technology</i>	2024 – 2028 CGPA: 7.8
BSF Senior Secondary School, Jammu <i>Class XII – PCM</i>	2024 86.8%
BSF Senior Secondary School, Jammu <i>Class X</i>	2020 94.6%

Technical Skills

Languages: Python, C++, C, Java, JavaScript, SQL
AI / Machine Learning: Generative AI, RAG, LLMs, Scikit-learn, TensorFlow, XGBoost, Pandas, NumPy
Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS
Backend: FastAPI, Node.js, Flask, REST APIs, JWT
Databases: PostgreSQL, SQLite, ChromaDB, Firebase
Tools / Platforms: Git, GitHub, Linux, Vercel, Render, Capacitor
Core Concepts: Data Structures, Algorithms, OOP, DBMS, Operating Systems, API Integration

Experience

Research Intern – IIT Jammu <i>NLP / Machine Learning Research</i>	2025
- Built a UMR parser for converting natural language into structured semantic graphs. - Developed Python pipelines for structured linguistic processing, data transformation, and NLP workflows. - Contributed to research-oriented workflows involving natural language processing and semantic representation.	
C++ Game Development Intern – INTERPE <i>Software Development</i>	Dec 2024 – Jan 2025
- Developed three console-based games using C++ with modular object-oriented design. - Applied OOP, STL, file handling, and core C++ programming concepts during development.	

Entrepreneurship

Founder & Full-Stack Developer – TutorNestJK <i>EdTech Platform</i>	Apr 2026 – Present <i>Jammu & Kashmir</i>
- Founded and independently developed a full-stack EdTech platform connecting students with tutors for personalized academic support. - Designed and implemented the platform website, user workflows, database integration, and deployment infrastructure. - Managed tutor onboarding, student coordination, product development, outreach, and platform operations. - Led product and growth initiatives from initial concept through deployment and operational execution.	

Projects

WhatsApp Study Bot – RAG-Powered AI Study Assistant WhatsApp API, Razorpay — GitHub	Python, FastAPI, Gemini, ChromaDB, SQLite,
- Built a WhatsApp-based RAG assistant that processes user notes and PDFs and generates context-grounded answers using Gemini and vector search. - Implemented per-user document isolation, usage tracking, and freemium subscription management using SQLite. - Integrated Meta WhatsApp Cloud API and Razorpay Payment Links with FastAPI webhook endpoints for an end-to-end AI application.	

PulseFit Neo – AI-Powered Fitness Platform Next.js, React, PostgreSQL, Drizzle, Capacitor, Clerk — GitHub

- Built a full-stack fitness platform with AI-generated personalized workout routines and real-time workout tracking.
- Integrated native device capabilities including accelerometer-based step tracking and GPS workout logging using Capacitor.
- Implemented secure authentication and a PostgreSQL-backed data layer using Clerk and Drizzle ORM.

Multi-Step Time Series Forecasting Python, TensorFlow, XGBoost — GitHub

- Built multi-step forecasting models to predict 7–30 future time steps using LSTM and XGBoost.
- Engineered lag features and rolling statistics to capture temporal patterns and improve forecasting performance.
- Compared tree-based and deep-learning approaches across multiple forecasting horizons.

ExpenseFlow – Finance Tracker React.js, Flask, SQLite, JWT — GitHub

- Developed a full-stack finance tracking platform with analytics-based expense management and spending visualization.
- Implemented JWT-based authentication and integrated React.js with Flask REST APIs for secure user workflows.
- Deployed the application using Vercel and Render.

Relevant Coursework

Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Machine Learning, Data Communication, Computer Organization and Architecture

Leadership & Achievements

- JK-CET 2024 – Rank 15
- Class Representative – NIT Srinagar
- SSB Conference Selected Candidate